Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/19/2024
PI Name:	Cromer, Walter Ph.D.		
Project Title:	The Effect of Simulated Space Radiation on the Interaction of the Metabolome, Immune System, and Lymphatic Anatomy of the Gastrointestinal Tract		
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Program/Discipline--Element/Subdiscipline:			
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Space Biology Special Category:	None		
PI Email:	wcromer@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	254-421-1530
Organization Name:	Texas A&M University		
PI Address 1:	Department of Medical Physiology		
PI Address 2:	2121 W. Holcombe		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	7
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Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
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COI Name (Institution):	Endsley, Mark Ph.D. (University of Texas, Galveston)		
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Task Description:	We will examine the changes that space relevant radiation have on the immune system, vascular architecture, and metabolome (bacterial products of metabolism) of the gastrointestinal (GI) tract. We know that each of these factors interact with each other to create a balanced system that ensures normal function of the digestive tract (digestion, nutrient absorption, protection from pathogens, tolerance of commensal bacteria) and promote the health of the whole organism. We believe that space radiation is the primary driving factor in imbalances between these factors which leads to system dysfunction. We will assess each of these factors (metabolome, immune, and vascular) in a manner that allows us to make conclusions about not only the state of each component individually but how it could impact the other systems. We will use standard metabolomic analysis of cecal contents (mass spectroscopy) accompanied by RNA deep sequencing to examine the pathways associated with those changes. This will allow us to determine if there are changes in metabolites that impact immune and GI function (indole, butyrate, etc.) and why. We will use immunofluorescent staining of sections of the bowel wall, Peyer's patches, and mesenteric lymph node to determine changes in the number and distribution of immune cells in the tissue. We will pair this with RNA sequencing of those tissues to determine the activation status of the cells and in the cases of the Peyer's patches and mesenteric nodes the production of factors (IL-7, CCL19, CCL21, etc.) by the stromovascular fraction that maintain a normal environment for the immune cells within. Finally, we will stain the aforementioned tissues for vascular markers (CD31, Lyve-1, etc.) to determine if there are changes in the vascular structures of the tissue. We will deliver a number of products from this proposal including the data listed as well as broad genomic and metabolomic data that will be archived for use by other investigators. More specifically we will provide data pertaining to degenerative effects of simulated space radiation on the interlocking systems of the metabolome, immune, and vascular system of the GI tract. The significance of this proposal is that it will be the first proposal to address the 3 listed components, metabolome, immune, and vasculature as a single system. We will also determine mechanistically how simulated space radiation interferes with each component individually and how that impacts the other linked systems.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	In addition to providing information on the effects of galactic cosmic radiation (GCR) simulated radiation this work will provide the first in depth analysis of the effect of radiation on lymphatic vessels. While there has been some work on radiation and lymph node and lymphatic endothelium there is no work on the effects on the conduit vessels. This will provide important information on the secondary effects of radiation therapy in association with lymphedema. We will also identify potential drug targets within the pathways that are effected.
Task Progress:	We have currently completed the tasks listed for this project. Due to budgetary constraints, primarily cost increases in the 2020-2022 timeframe, we were limited in the analysis that we were able to perform on female tissues. The outcomes are as follows: RNA Sequencing and Analysis: Currently, we have completed all attempts of RNA deep sequencing of tissues associated with the GI tract. There were failures to extract sufficient levels of RNA from mesenteric lymphatic vessels for use in the proposed screening methods. All other tissues were extracted and analyzed successfully. These data were fed into an analysis pipeline built by a colleague at the Texas A&M IBT. From this analysis there were several unexpected findings; first, the changes in RNA expression appear to be radiation type agnostic using unsupervised clustering (between GCR sim and gamma) (figure 1a) though when corrected for tissue type (guided clustering) there was a treatment effect (radiation type clustering) (figure 1b); unique gene expression patterns outside of what is expected with heavy co-alignment with metabolic diseases (the expected vascular and cancer fingerprints were also present) (figure 1c, file:///private/var/folders/0j/7xdq3y850y91mcskrl8kp6yw0000gr/T/com.microsoft.Outlook/Outlook%20Temp/DEG_term_enrichment_network%5B98%5D.html); gene profiles favored downregulation of genes (we expected an equal up and down regulation) (figure 1d). These data are only a partial picture as these data only come from males due to delays associated with the supply chain and increases in prices, we were unable to perform female sequencing (12 months from isolation to sequencing and an increase of ~75% in cost). Though from metabolic analysis there is a need to perform the female sequencing which we are finding funds to do so from outside sources currently. We chose to examine males first due to susceptibility to forms of colorectal cancer (CRC) and gamma irradiated males did show a closer pattern matching to fingerprints associated with azoxymethane + dextran sodium sulfate (AOM+DSS) model of CRC. Interestingly, the GCR irradiated animals had a randomly dispersed pattern of expression that did not match CRC though there were matches to other types of cancer (figure 2a). There was a strong link in both radiation types and all tissues with cancers that result from Smad4 and Pik3r2 gene errors (figure 1c file:///private/var/folders/0j/7xdq3y850y91mcskrl8kp6yw0000gr/T/com.microsoft.Outlook/Outlook%20Temp/DEG_term_enrichment_network%5B98%5D.html). The genes we had identified as critical to maintaining normal immune homeostasis within the lymph node were also changed but not in the manner we suspected (figure 2c). We presumed that the genes that maintain homeostasis would be decreased by radiation as injury to the cells of the stromovascular fraction reduced their viability. This is not the case and changes were almost universally in the positive direction. The increase in both regulatory genes, used here as genes that prevent aberrant activation of cells in the node, and gene associated with immune cell activation suggest an impaired coordination of the adaptive immune response. This may explain some of the findings in the literature pointing at this. Of particular interest was evidence from more in-depth bioinformatic analysis that the pathways associated with antigen processing and presentation in all tissues were disrupted (figure 1c file:///private/var/folders/0j/7xdq3y850y91mcskrl8kp6yw0000gr/T/com.microsoft.Outlook/Outlook%20Temp/DEG_term_enrichment_network%5B98%5D.html) providing more evidence for changes in the adaptive immune response. This is linked with signs of different types of infections, both viral and bacterial, in the animals is even stronger evidence for disrupted immunity in these tissues due to radiation. While these are not novel findings, there is evidence in the literature supporting both carcinogenesis and immune suppression due to radiation going back many decades, this is a pseudo-mechanistic explanation for these changes and the most detailed genomic analysis of these changes in GCR sim to date. Building on this, we have built a bioinformatics pipeline that can predict which drug or drug combinations can reverse these gene changes and prevent the potential disruptions these gene changes could bring about. We have identified 2 broad categories of drugs that may revert these changes, HDAC inhibitors and multi-kinase inhibitors. We are in the process of developing methods to test these drugs in relevant models (high-throughput screening based on cancer drug screens run at TAMU IBT) (figure 3). Metabolomic Analysis: Again, we had to make a compromise in our analysis as we only had enough sample/ funds to perform either metabolomic analysis or microbiomic analysis. We chose to focus on metabolomic analysis due to the abundance of investigators performing microbiomic analysis, our previous data showing little variation in microbiomic composition in well controlled spaceflight, the major shifts in the metabolome of spaceflight independent of similar changes in the microbiome, the relative lack of metabolomic data available for both spaceflight and the relevant radiation types, and the fact that the metabolome performs a vital “endocrine” function in the host. The results of the short chain fatty acid analysis showed something that we did not expect, there was a stark difference in the metabolomic responses in males and females (figure 4) that appeared to be radiation type agnostic. This was particularly apparent in butyric, propionic, and valeric acids all of which are critical to host immune processes. This is very interesting as butyric acid is a key regulator of HDAC expression and activity which was identified as a target in our bioinformatics analysis. Also, it appears the female response to the radiation types more closely matches what is seen in spaceflight. The caveat to this is that the spaceflight data was generated in males as the study in which we had access to the pertinent samples only included males in the study design. We are hoping to add spaceflight data from females to see if this trend continues. Cytokine Analysis: We utilized z-score data from the tissues to perform this analysis. This is not the most ideal method as there can be situations where transcript levels do not match translation. This is, however, a very minor concern. Our data from tissues have shown in the past to be most affected by spaceflight (in terms of cytokine production) revealed no significant shifts in cytokine transcript levels. This fits with the overall conclusion that while the changes we have seen may prime the system for failure, without a challenge, the end stage effectors remain unchanged.

Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Cromer W. "Effect of galactic cosmic radiation simulation on intestinal and associated lymphoid tissues. " 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Cromer W. "Space type radiation creates a specific pharmacogenetic fingerprint in intestinal tissue. " 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Cromer, W. "Low Dose Space Relevant Radiation Causes a Unique Transcriptomic Phenotype in Gut associated Lymphoid Organs " 2024 Lymphatic Forum Banff, Alberta, Canada, 2024. Abstracts. Lymphatic Forum, Banff, Alberta, Canada, 2024. , May-2024